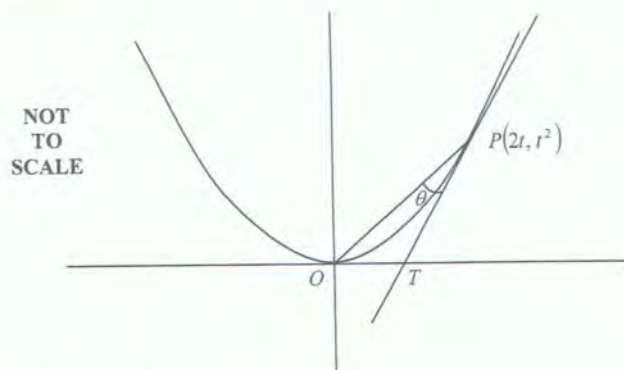


Marks

Question 3 (12 marks) Use a SEPARATE writing booklet.

- (a) $P(2t, t^2)$ is a point on the parabola $x^2 = 4y$ where $t > 0$.
The tangent to the parabola at P cuts the x -axis at T .
 $\angle OPT = \theta$



- (i) Find the gradients of OP and PT . 2
- (ii) Show that $\tan \theta = \frac{t}{t^2 + 2}$. 2
- (b) (i) Show that the function $f(x) = \frac{x-4}{x-2}$, $x \neq 2$ is increasing for all values of x in its domain. 1
- (ii) Sketch the graph of the function, showing clearly the coordinates of any points of intersection with the x -axis and y -axis, and the equations of any asymptotes. 2
- (iii) Find the inverse function, $f^{-1}(x)$, and state its range. 2

Question 3 continues on page 5

Marks

Question 3 (continued)

- (c) Let each different arrangement of all the letters of $GOOGLEPLEX$ be called a word.
- (i) How many words are possible? 1
- (ii) If one of these words is chosen at random, what is the probability that all the vowels are together? 2

End of Question 3

Question 4 (12 marks) Use a SEPARATE writing booklet.

Marks

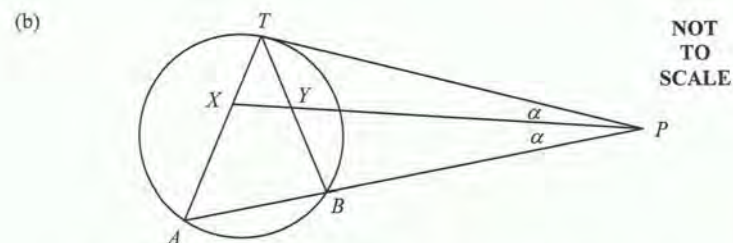
- (a) Find the term independent of x in the expression $\left(\frac{1}{3x} - \frac{3}{2}x^2\right)^9$. 3
- (b) Given that a root of the equation $e^x - x - 2 = 0$ is close to $x = 1.2$, use one application of Newton's Method of Approximation to find a second approximation of this root. (Answer to 2 decimal places.) 3
- (c) Find the exact volume of the solid formed when the region bounded by the x -axis and the curve $y = x(8 - x^3)^3$ between $x = 0$ and $x = 2$ is rotated about the x -axis. (You may need to use the substitution $u = 8 - x^3$.) 3
- (d) At a factory that produces Ipads it was found that on average 5% of Ipads produced have a fault. A batch of 15 Ipads is tested.
- (i) What is the probability that there are exactly 2 Ipads containing faults in the batch of 15? (Answer to 2 decimal places.) 1
- (ii) What is the probability that at least 1 Ipad contains a fault? (Answer to 2 decimal places.) 2

End of Question 4

Question 5 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) On old 727 jet planes there are 56 rows of seats. Each row has three seats on each side of a central aisle. Three friends took a flight on a 727 jet plane with random seat allocation. Find the number of seating arrangements possible for the three friends if:
- (i) all three friends must sit together on one side of the aisle. 1
- (ii) all three must sit in separate rows. 1
- (iii) no more than two can sit together. 1



In the diagram above the tangent at T on the circle meets a chord AB produced to P . The bisector of $\angle TPA$ meets TA and TB at X and Y respectively.

- (i) Give a reason why $\angle PTB = \angle TAB$. 1
- (ii) Prove $TX = TY$. 2
- (iii) Prove $\frac{TX}{AX} = \frac{TP}{AP}$. 2

Question 5 continues on page 8

Question 5 (continued)

Marks

- (c) (i) Differentiate $y = \tan^{-1} \frac{1}{x}$, $x \neq 0$ and hence show that

2

$$\frac{d}{dx} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) = 0$$

- (ii) Sketch the curve $y = \tan^{-1} x + \tan^{-1} \frac{1}{x}$

2

End of Question 5

Marks

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the binomial expansion

$$1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n = (1+x)^n$$

- (i) Show that $1 - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0$

1

- (ii) Show that $1 - \frac{1}{2}\binom{n}{1} + \frac{1}{3}\binom{n}{2} - \dots + (-1)^n \frac{1}{n+1}\binom{n}{n} = \frac{1}{n+1}$

3

- (b) $P(2at, at^2)$ is a point on the parabola $x^2 = 4ay$ with focus $S(0, a)$.

- (i) Derive the equation of the normal to the parabola at P .

2

- (ii) The normal meets the y -axis at G . Show that the coordinates of G are $(0, 2a + at^2)$.

1

- (iii) Find the length of GP and PS in terms of a and t .

2

- (iv) Given that $\triangle SPG$ is equilateral, prove there are two positions of P and give the coordinates of these points in terms of a .

3

End of Question 6

Marks

Question 7 (12 marks) Use a SEPARATE writing booklet.

(a) Suppose $(5 + 2x)^{12} = \sum_{k=0}^{12} t_k x^k$.

(i) Use the binomial theorem to write an expression for t_k where $0 \leq k \leq 12$. 1

(ii) Show that $\frac{t_{k+1}}{t_k} = \frac{2(12-k)}{5(k+1)}$. 1

(iii) Hence or otherwise, find the greatest coefficient. 2

(b) Find the general solution to $\sin 2\theta + \sqrt{3} \cos 2\theta = 0$. 2

(c) (i) Show that $\sin A + \cos A = \sqrt{2} \sin\left(A + \frac{\pi}{4}\right)$. 2

(ii) Prove that the derivative of $y = e^x \sin x$ is given by $\frac{dy}{dx} = \sqrt{2} e^x \sin\left(x + \frac{\pi}{4}\right)$. 1

(iii) Given the function $y = e^x \sin x$, prove by mathematical induction that the n th derivative of y for a positive integer n is: 3

$$\frac{d^n y}{dx^n} = (\sqrt{2})^n e^x \sin\left(x + \frac{n\pi}{4}\right)$$

$$\left(\text{Note: } \frac{d}{dx} \left(\frac{d^k y}{dx^k}\right) = \frac{d^{k+1} y}{dx^{k+1}}\right)$$

End of paper

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Question 1

a) $\lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{5}{2}$

$$= 1 \times \frac{5}{2}$$

$$= \frac{5}{2}$$

b) $x(2x-3) \geq x^2$
 $2x^2 - 3x \geq x^2$
 $x^2 - 3x \geq 0$
 $x(x-3) \geq 0$



(R
3)

(Com
3)

$\therefore x \leq 0, x \geq 3$ but $x \neq 0$

$\therefore$ solution is $x < 0, x \geq 3$

c) $A(4, 3)$ $B(1, -1)$
 $-3 : 2$

$$x = \frac{-3 \times 1 + 2 \times 4}{-3 + 2}$$

$$x = -5$$

$$y = \frac{-3 \times -1 + 2 \times 3}{-3 + 2}$$

$$y = -9$$

d) $\int_0^1 \frac{dx}{\sqrt{2-x^2}} = \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1$

$$= \sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} 0$$

$$= \frac{\pi}{4}$$

(Ca
2)

e) i) Show $\tan x = \frac{\sin 2x}{1 + \cos 2x}$

$$RHS = \frac{2 \sin x \cos x}{1 + 2 \cos^2 x - 1}$$

$$= \frac{2 \sin x}{2 \cos x}$$

$$= \tan x$$

$$= LHS$$

(R
2)

ii) $\tan \frac{\pi}{12} = \frac{\sin \frac{\pi}{6}}{1 + \cos \frac{\pi}{6}}$

$$= \frac{1}{2} \div \left(1 + \frac{\sqrt{3}}{2} \right)$$

$$= \frac{1}{2} \times \frac{2}{2 + \sqrt{3}}$$

$$= \frac{1}{2 + \sqrt{3}}$$

Question 2.

a) $\angle BOD = 2\angle BAD$ ($\angle$ at the centre is twice the angle at the circumference on the same arc) ✓
 $= 2x$
 $\angle BCD = \angle BOD$ (given)
 $= 2x$

$\angle BAD + \angle BCD = 180^\circ$ (opposite angles in a cyclic quad are supplementary) ✓
 $x + 2x = 180^\circ$
 $3x = 180^\circ$
 $x = 60^\circ$

b) $P(-1) = -a + b + 19 - 15$ $P(3) = 27a + 9b - 57 - 15$
 $0 = -a + b + 4$ $0 = 27a + 9b - 72$
 $a = b + 4$ ① ✓ $0 = 3a + b - 8$ ② ✓

sub ① into ②

$$0 = 3(b+4) + b - 8$$

$$0 = 3b + 12 + b - 8$$

$$4b = -4$$

$$b = -1 \therefore a = 3$$
 ✓

c) $3^{2n} - 1$
for $n = 1$
 $3^{2n} - 1 = 9 - 1$ ✓
 $= 8$ which is divisible by 8 $\therefore$ true for $n = 1$

assume true for $n = k$

ie $3^{2k} - 1 = 8p$ (p being some positive integer)
 $3^{2k} = 8p + 1$

RTP for $n = k + 1$

$$3^{2n} - 1 = 3^{2k+2} - 1$$

$$= 3^{2k} \times 3^2 - 1$$

$$= (8p+1) \times 9 - 1$$
 (using assumption) ✓

$$= 72p + 9 - 1$$

$$= 72p + 8$$

$$= 8(9p+1)$$
 ✓

divisible by 8 since p is an integer

true for $n = k$ and for $n = k + 1$, since also true for $n = 1$ it follows that it is true for $n = 2, 3$ etc $\therefore$ true for all positive integers ✓

d) $u = x + 1$ $\int \frac{3x+1}{\sqrt{x+1}} dx$
 $\frac{du}{dx} = 1$
 $du = dx$ $= \int \frac{3(u-1)+1}{u^{\frac{1}{2}}} du$ ✓
 $= \int \frac{3u-2}{u^{\frac{1}{2}}} du$
 $= \int 3u^{\frac{1}{2}} - 2u^{-\frac{1}{2}} du$ ✓
 $= \frac{2}{3} \times 3u^{\frac{3}{2}} - \frac{2}{1} \times 2u^{\frac{1}{2}}$
 $= 2u^{\frac{3}{2}} - 4u^{\frac{1}{2}}$ ✓
 $= 2u^{\frac{1}{2}}(u-2)$
 $= 2\sqrt{x+1}(x-1) + C$

Question 3

a) $P(2t, t^2)$ $x^2 = 4y$ $t > 0$

i) $M_{OP} = \frac{t^2}{2t}$ $M_{PT} = \frac{dy}{dx}$
 $= \frac{t}{2}$ $= \frac{2x}{4}$
 $= \frac{x}{2}$ at $x=2t$
 $= t$

ii) $\angle$ between two lines: $\tan \theta = \left| \frac{m_2 - m_1}{1 + m_2 m_1} \right|$

$\therefore \tan \theta = \frac{t - \frac{t}{2}}{1 + t \times \frac{t}{2}}$ $t > 0$
 $= \frac{2t - t}{2} \div \frac{2 + t^2}{2}$
 $= \frac{2t - t}{2 + t^2}$
 $= \frac{t}{t^2 + 2}$

b) i) $f(x) = \frac{x-4}{x-2}$, $x \neq 2$

$f'(x) = \frac{(x-2) - (x-4)}{(x-2)^2}$ $u = x-4$ $v = x-2$
 $u' = 1$ $v' = 1$

$f'(x) = \frac{2}{(x-2)^2}$

since $\frac{2}{(x-2)^2} > 0$ $f'(x) > 0 \therefore f(x)$ is increasing

ii) points of intersection: x intercept = 4
 y intercept = 2

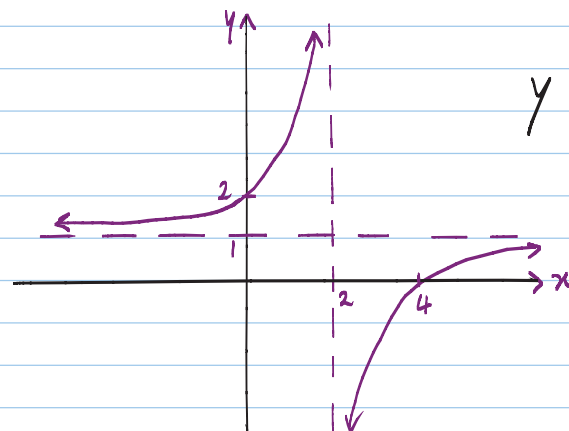
asymptotes: vertical when $x = 2$

horizontal when $x \rightarrow \infty$

$\lim_{x \rightarrow \infty} \frac{\frac{x}{x} - \frac{4}{x}}{\frac{x}{x} - \frac{2}{x}}$

$= \frac{1-0}{1-0}$

$y = 1$



iii) $f^{-1}(x)$: $x = \frac{y-4}{y-2}$

$xy - 2x = y - 4$

$xy - y = 2x - 4$

$y(x-1) = 2(x-2)$

$y = \frac{2(x-2)}{x-1}$

Range: all real y , $y \neq 2$

$$c) \left. \begin{array}{l} C \times 2 \\ O \times 2 \\ L \times 2 \\ E \times 2 \\ P \\ X \end{array} \right\} 10 \text{ letters}$$

$$i) \frac{10!}{2 \times 2 \times 2 \times 2} = 226800$$

$$ii) \frac{7!}{2 \times 2} \times \frac{4!}{2 \times 2}$$

$$= 1260 \times 6$$

$$= 7560$$

Question 4

$$a) \left(\frac{1}{3x} - \frac{3}{2}x^2 \right)^9$$

$$\text{General Term: } \binom{9}{k} \left(\frac{1}{3}x^{-1} \right)^{9-k} \left(-\frac{3}{2}x^2 \right)^k \quad \checkmark$$

$$\text{Find } k: x^{-9+k} \times x^{2k} = x^0$$

$$-9 + k + 2k = 0$$

$$3k = 9$$

$$k = 3 \quad \checkmark$$

$$\text{Independent Term: } \binom{9}{3} \left(\frac{1}{3}x^{-1} \right)^6 \left(-\frac{3}{2}x^2 \right)^3$$

$$= 84 \times \frac{1}{729} \times -\frac{27}{8}$$

$$= -\frac{7}{18} \quad \checkmark$$

$$b) \begin{array}{l} e^x - x - 2 = 0 \quad x = 1.2 \\ f(1.2) = e^{1.2} - 1.2 - 2 \\ = 0.1201 \end{array}$$

$$\begin{array}{l} f'(x) = e^x - 1 \\ f'(1.2) = e^{1.2} - 1 \\ = 2.3201 \end{array}$$

$$x_1 = 1.2 - \frac{0.1201}{2.3201}$$

$$= 1.15$$

$$c) V = \pi \int_0^2 y^2 dx$$

$$= \pi \int_0^2 x^2 (8-x^3)^6 dx$$

$$= \pi \int_8^0 u^6 x^2 \times \frac{du}{-3x^2}$$

$$= -\frac{\pi}{3} \int_8^0 u^6 du \quad \checkmark$$

$$y = x(8-x^3)^3$$

$$y^2 = x^2(8-x^3)^6$$

$$u = 8 - x^3$$

$$\frac{du}{dx} = -3x^2$$

$$dx = \frac{du}{-3x^2} \quad \checkmark$$

$$\text{when } x = 0, u = 8$$

$$x = 2, u = 0$$

$$= -\frac{\pi}{3} \left[\frac{u^2}{7} \right]_8^0$$

$$= -\frac{\pi}{3} \left(0 - \frac{8^2}{7} \right)$$

$$= \frac{8^2 \pi}{21} \quad \text{or} \quad \frac{2097152\pi}{21}$$

d) Good Faulty $n=15$
 $0.95 \quad 0.05$

i) 2 faults = $\binom{15}{2} 0.95^{13} \times 0.05^2$
 $= 0.13$

ii) at least 1 fault = $1 - \text{no faults}$
 $= 1 - \binom{15}{0} 0.95^{15}$
 $= 0.54$

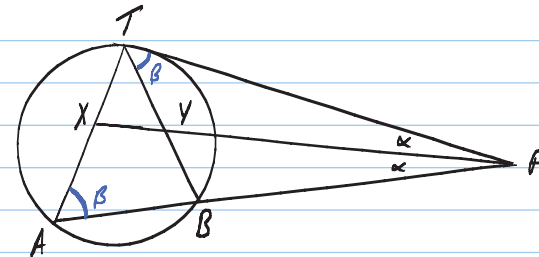
Question 5

a) $56 \times 6 = 336$ seats

i) $336 \times 2 \times 1 = 672$

ii) $336 \times 330 \times 324 = 35925120$

iii) no more than 2 together = no restriction - 3 together
 $= (336 \times 335 \times 334 - 672)$
 $= 37594368$



b) i) $\angle PTB = \angle TAB$ (angle between a tangent and chord is equal to the angle in the alternate segment)

ii) let $\angle PTB = \angle TAB = \beta$
 $\angle TXP = \alpha + \beta$ (exterior $\angle$ of Δ = sum opp interior $\angle$ s)
 $\angle TYX = \alpha + \beta$ (" " " " ")
 $\therefore \angle TXP = \angle TYX \therefore \Delta TXY$ is isosceles
 $TX = TY$ (sides opp = angles in an isosceles Δ are =)

iii) in ΔTYP and ΔAXP
 $\angle TPY = \angle APX$ (given)
 $\angle PTY = \angle PAX$ (proven in (i))
 $\therefore \Delta TYP \parallel \Delta AXP$ (equiangular)
 since corresponding sides in similar Δ s are in proportion
 $\frac{TX}{AX} = \frac{TP}{AP}$

c) i) $y = \tan^{-1} \frac{1}{x}$

$y = \tan^{-1} x^{-1}$

$$\frac{dy}{dx} = \frac{-x^{-2}}{1+x^{-2}}$$

$$= -\frac{1}{x^2} \times \frac{1}{\frac{x^2+1}{x^2}}$$

$$= -\frac{1}{x^2} \times \frac{x^2}{x^2+1}$$

$$= -\frac{1}{x^2+1}$$

also if $y = \tan^{-1} x$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

(P2)

$$\frac{d}{dx} \left(\tan^{-1} x + \tan^{-1} \frac{1}{x} \right) = \frac{d}{dx} (\tan^{-1} x) + \frac{d}{dx} \left(\tan^{-1} \frac{1}{x} \right)$$

$$= \frac{1}{1+x^2} - \frac{1}{1+x^2}$$

$$= 0$$

ii) if $\frac{d}{dx} = 0$ the curve is horizontal for all x in the domain

also $x \neq 0$

test a point when $x > 0$

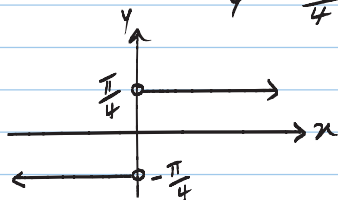
ie let $x = 1$ $y = \tan^{-1} 1$

$$y = \frac{\pi}{4}$$

test a point when $\frac{x}{x} < 0$

$$y = \tan^{-1} (-1)$$

$$y = -\frac{\pi}{4}$$



Com 2

Question 6

a) $1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n = (1+x)^n$

i) substitute $x = -1$

$$1 + -\binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + \binom{n}{n}(-1)^n = (1-1)^n$$

$$1 - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots + (-1)^n \binom{n}{n} = 0$$

ii) Integrate original

$$x + \frac{1}{2} \binom{n}{1} x^2 + \frac{1}{3} \binom{n}{2} x^3 + \dots + \frac{1}{n+1} \binom{n}{n} x^{n+1} = \frac{1}{n+1} (1+x)^{n+1} + C$$

let $x = 0$, $0 = \frac{1}{n+1} + C$

$$C = -\frac{1}{n+1}$$

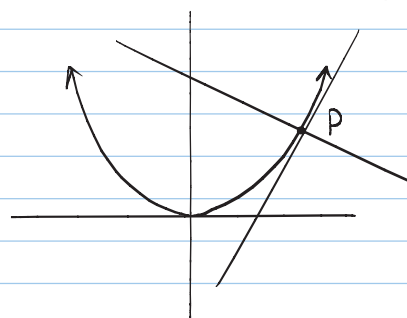
sub $x = -1$

$$-1 + \frac{1}{2} \binom{n}{1} - \frac{1}{3} \binom{n}{2} + \dots + \frac{1}{n+1} \binom{n}{n} (-1)^{n+1} = 0 - \frac{1}{n+1}$$

$$-1 + \frac{1}{2} \binom{n}{1} - \frac{1}{3} \binom{n}{2} + \dots + \frac{1}{n+1} \binom{n}{n} (-1)^n (-1) = -\frac{1}{n+1}$$

$$1 - \frac{1}{2} \binom{n}{1} + \frac{1}{3} \binom{n}{2} - \dots + (-1)^n \frac{1}{n+1} \binom{n}{n} = \frac{1}{n+1}$$

b) $P(2at, at^2)$ $x^2 = 4ay$ $S(0, a)$



i) $y = \frac{x^2}{4a}$

$$y' = \frac{x}{2a}$$

$$M_T = t \quad M_N = -\frac{1}{t}$$

Egn: $y - at^2 = -\frac{1}{t}(x - 2at)$

$$yt - at^3 = -x + 2at$$

$$x + yt = 2at + at^3$$

ii) G lies on y-axis $\therefore x=0$

$$0 + yt = 2at + at^3$$

$$y = 2a + at^2 \quad \therefore G(0, 2a + at^2)$$

iii) $GP = \sqrt{(2at)^2 + (at^2 - 2a - at^2)^2}$

$$= \sqrt{4a^2t^2 + 4a^2}$$

$$= 2a\sqrt{t^2 + 1}$$

$$PS = \sqrt{(2at)^2 + (at^2 - a)^2}$$

$$= \sqrt{4a^2t^2 + a^2t^4 - 2a^2t^2 + a^2}$$

$$= a\sqrt{2t^2 + t^4 + 1}$$

$$= a\sqrt{(t^2 + 1)^2}$$

$$= a(t^2 + 1)$$

$$(t^2 + 1)^2$$

iv) If $\triangle SPG$ is equilateral $GP = PS$

$$2a\sqrt{t^2 + 1} = a(t^2 + 1)$$

$$4(t^2 + 1) = (t^2 + 1)^2$$

$$0 = (t^2 + 1)^2 - 4(t^2 + 1)$$

$$0 = (t^2 + 1)(t^2 + 1 - 4)$$

$$0 = (t^2 + 1)(t^2 - 3)$$

$$\therefore t^2 = -1, \quad t^2 = 3$$

no soln.

$$t = \pm\sqrt{3}$$

$$\therefore \text{point } (2\sqrt{3}a, 3a) \text{ or } (-2\sqrt{3}a, 3a)$$

Question 7.

a) $(5+2x)^{12} = \sum_{k=0}^{12} T_k$

i) $T_k = \binom{12}{k} 5^{12-k} 2^k x^k$

ii) $T_{k+1} = \binom{12}{k+1} 5^{11-k} 2^{k+1} x^{k+1}$

$$\frac{T_{k+1}}{T_k} = \left(\frac{12!}{(k+1)!(11-k)!} \times 5^{11-k} \times 2^{k+1} \right) \div \left(\frac{12!}{k!(12-k)!} \times 5^{12-k} \times 2^k \right)$$

$$= \frac{12!}{(k+1)k!(11-k)!} \times \frac{k!(12-k)!(11-k)!}{12!} \times 5^{-1} \times 2^1$$

$$= \frac{2(12-k)}{5(k+1)}$$

ii) $\frac{2(12-k)}{5(k+1)} \geq 1 \quad 0 \leq k \leq 12$

$$24 - 2k \geq 5k + 5$$

$$19 \geq 7k$$

$$2.7 \geq k$$

$$\therefore k = 2$$

$$\text{greatest coefficient} = \binom{12}{3} 5^9 2^3$$

$$(T_{k+1}) = 3\,437\,500\,000$$

(R
2)

b) $\sin 2\theta + \sqrt{3} \cos 2\theta = 0$

$$\tan 2\theta + \sqrt{3} = 0$$

$$\tan 2\theta = -\sqrt{3} \quad Q2, Q4$$

$$\text{related angle} = \frac{\pi}{3}$$

$$Q2: 2\theta = \pi - \frac{\pi}{3}, 2\pi + \pi - \frac{\pi}{3}, 4\pi + \pi - \frac{\pi}{3} \dots$$

$$Q4: 2\theta = 2\pi - \frac{\pi}{3}, 2\pi + 2\pi - \frac{\pi}{3}, 4\pi + 2\pi - \frac{\pi}{3}$$

$$\text{General Soln: } 2\theta = n\pi - \frac{\pi}{3} \quad \checkmark$$

$$\theta = \frac{n}{2}\pi - \frac{\pi}{6} \quad \checkmark$$

$$\left[\begin{array}{l} \text{OR by formula } 2\theta = n\pi + \tan^{-1}(-\sqrt{3}) \\ 2\theta = n\pi - \frac{\pi}{3} \\ \theta = \frac{n}{2}\pi - \frac{\pi}{6} \end{array} \right]$$

$$d) i) \sin A + \cos A = \sqrt{2} \sin\left(A + \frac{\pi}{4}\right)$$

$$\text{RHS} = \sqrt{2} \left(\sin A \cos \frac{\pi}{4} + \sin \frac{\pi}{4} \cos A \right)$$

$$= \sqrt{2} \left(\sin A \times \frac{1}{\sqrt{2}} + \cos A \times \frac{1}{\sqrt{2}} \right)$$

$$= \sin A + \cos A$$

$$= \text{LHS}$$

$$ii) y = e^x \sin x \quad u = e^x \quad v = \sin x \\ u' = e^x \quad v' = \cos x$$

$$\frac{dy}{dx} = e^x \sin x + e^x \cos x$$

$$= e^x (\sin x + \cos x)$$

$$= e^x \left[\sqrt{2} \sin\left(x + \frac{\pi}{4}\right) \right] \quad (\text{from (i)})$$

$$= \sqrt{2} e^x \sin\left(x + \frac{\pi}{4}\right)$$

$$iii) y = e^x \sin x \quad \frac{d^n y}{dx^n} = \sqrt{2}^n e^x \sin\left(x + \frac{\pi}{4}\right)$$

• for $n=1$

$$\frac{dy}{dx} = \sqrt{2} e^x \sin\left(x + \frac{\pi}{4}\right)$$

true as proven in (ii)

(R3)

• assume true for $n=k$

$$\frac{d^k y}{dx^k} = \sqrt{2}^k e^x \sin\left(x + \frac{k\pi}{4}\right)$$

• RTP for $n=k+1$

$$\frac{d^{k+1} y}{dx^{k+1}} = \sqrt{2}^{k+1} e^x \sin\left(x + \frac{\pi(k+1)}{4}\right)$$

$$\text{since } \frac{d^{k+1} y}{dx^{k+1}} = \frac{d}{dx} \left(\frac{d^k y}{dx^k} \right)$$

from assumption

$$\frac{d}{dx} \left(\frac{d^k y}{dx^k} \right) = \frac{d}{dx} \left[\sqrt{2}^k e^x \sin\left(x + \frac{k\pi}{4}\right) \right] \quad \begin{array}{l} u = e^x \quad v = \sin\left(x + \frac{k\pi}{4}\right) \\ u' = e^x \quad v' = \cos\left(x + \frac{k\pi}{4}\right) \end{array}$$

$$= \sqrt{2}^k \left[e^x \sin\left(x + \frac{k\pi}{4}\right) + e^x \cos\left(x + \frac{k\pi}{4}\right) \right]$$

$$= \sqrt{2}^k e^x \left[\sin\left(x + \frac{k\pi}{4}\right) + \cos\left(x + \frac{k\pi}{4}\right) \right]$$

from (ii)

$$= \sqrt{2}^k e^x \left[\sqrt{2} \sin\left(x + \frac{k\pi}{4} + \frac{\pi}{4}\right) \right]$$

$$= \sqrt{2}^{k+1} e^x \sin\left(x + \frac{(k+1)\pi}{4}\right)$$